Reading Between the Lines: How People Perceive and Detect Cultural Erasure in AI-Retold Bengali Folktales

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Abstract

Large language models (LLMs) are becoming everyday interfaces through which people encounter, rewrite, translate, and circulate cultural narratives. This paper examines whether readers can perceive cultural erasure when Bengali folktales are retold by contemporary generative AI systems. We present a mixed-methods study of 27 Bengali folktales, 1,296 AI-generated retellings, and a human verification study with 76 participants. We first construct a bilingual reference corpus annotated for culturally non-substitutable lexical items, atomic plot points, socio-cultural behavioral traits, and imposed Western narrative forms. We then introduce the Cultural Fidelity Perception Framework, a rubric-based audit pipeline that moves beyond lexical overlap and embedding similarity to evaluate deeper cultural displacement. The framework includes the Structural Displacement Indicator (SDI), which captures cases where a retelling preserves surface coherence while replacing culturally situated narrative structure with generic or Westernized story grammar. Across six LLMs and four prompt strategies, retellings showed systematic cultural compression, with Bengali-mediated interaction producing especially sharp losses in cultural retention. In a human study, participants performed near-chance when distinguishing high-fidelity AI-generated retellings from authentic stories under blind-reading conditions, but detection improved when distortions were severe or when automated feedback scaffolded attention to cultural cues. We discuss six HCI design implications for culturally accountable generative systems: separating fluency from cultural fidelity, exposing language-of-interaction asymmetries, defaulting to source-text anchoring, calibrating friction for cultural risk, supporting community validation, and detecting structural displacement at scale.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; • **Computing methodologies** → *Natural language processing*.

Keywords

Human-computer interaction, cultural erasure, Bengali folktales, large language models, cultural fidelity, human-AI evaluation

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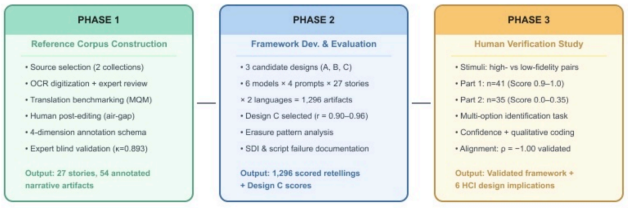


Figure 1: Overview of the three-phase study design: reference corpus construction, automated cultural fidelity auditing, and human verification.

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1 Introduction

Generative AI systems increasingly mediate how cultural stories are summarized, translated, rewritten, and shared. In many everyday settings, users do not encounter a cultural source directly; they encounter an AI-generated retelling that may be more fluent, more familiar, or more legible to global audiences than the original. This mediation matters for low-resource and underrepresented cultures because cultural loss is not always visible as factual error. A story can preserve names, characters, and plot sequence while replacing local social relationships, moral ambiguity, ritual vocabulary, or cosmological assumptions with a more generic narrative form.

Bengali folktales provide a useful site for studying this problem. They are culturally dense, often orally transmitted, and frequently depend on social roles, kinship terms, local settings, supernatural conventions, and narrative structures that do not map cleanly onto dominant Western storytelling patterns. When such stories are retold by LLMs, cultural erasure may take the form of lexical substitution, plot simplification, behavioral normalization, or the insertion of universalized morals. These transformations can be difficult to detect because the resulting text may remain fluent and coherent.

This paper asks: how do LLMs transform Bengali folktales, and when can human readers detect those transformations? We address this question through the three-phase research design shown in Figure 1. First, we construct an annotated reference corpus of 27 Bengali folktales represented as 54 Bengali-English story pairs. Second, we generate and audit 1,296 AI-generated retellings from six contemporary LLMs under four prompt strategies. Third, we conduct a human verification study with 76 participants to compare automated cultural fidelity scores with readers' detection behavior.

A central challenge for HCI is that cultural erasure is often experienced through interaction rather than through a single incorrect sentence. Users ask a system to make a story shorter, clearer, more child-friendly, or easier to translate. Each request appears benign, yet the cumulative effect can be the removal of culturally specific details that make the narrative recognizable to its community. In Bengali folktales, such details include forms of address, kinship obligations, local material culture, supernatural causality, and the social consequences of cleverness, greed, loyalty, or disobedience. If an interface presents the resulting story as simply an improved version of the source, users may have little reason to inspect what was lost.

We use the term cultural fidelity to refer to the degree to which a retelling preserves culturally meaningful elements of a source narrative. Cultural fidelity is not identical to literal translation, factual accuracy, or semantic similarity. A faithful retelling may change wording while preserving narrative relationships and cultural logic; an unfaithful retelling may keep the plot outline while replacing the cultural frame. This distinction is important for generative AI evaluation because many automated metrics reward surface similarity or fluent paraphrase rather than preservation of locally grounded meaning.

We make three contributions to HCI and human-centered AI. First, we operationalize cultural fidelity for Bengali folktale retelling through a bilingual reference corpus annotated across lexical, structural, behavioral, and narrative-form dimensions. Second, we introduce the Cultural Fidelity Perception Framework and SDI as tools for identifying deeper cultural displacement that conventional lexical or embedding-based metrics miss. Third, we provide evidence from human participants that unaided readers may be poorly calibrated when AI outputs are fluent and culturally plausible, motivating interface-level scaffolds that make cultural fidelity visible.

2 Related Work

2.1 Cultural Bias and Multicultural LLM Evaluation

Recent work has shown that LLMs vary across languages, cultures, and geopolitical perspectives, raising concerns about whose norms are centered in generated text [4, 7]. Multicultural evaluation benchmarks have expanded beyond English-centric tasks, but many still focus on question answering, factual knowledge, or value alignment rather than narrative transformation [8, 9]. Bengali folktales require a different evaluation lens because fidelity depends not only on whether the model remembers a fact, but also on whether it preserves situated meaning across retelling.

2.2 AI Writing, Homogenization, and Cultural Nuance

HCI research on AI-assisted writing has shown that automated suggestions can improve perceived fluency while narrowing stylistic diversity. Agarwal et al. argue that AI suggestions can homogenize writing toward Western styles and reduce cultural nuance [1]. Folktale retelling makes this risk concrete: LLMs may transform ambiguous, locally grounded, or collectively oriented narratives into stories with individual heroic arcs, explicit moral lessons, or simplified emotional closure. Such outputs are not necessarily low quality by generic readability standards, but they can be culturally unfaithful.

2.3 Digital Cultural Heritage and Human-AI Collaboration

Cultural heritage preservation involves interpretive authority, not only digitization. Policy and digital heritage work emphasizes that communities must retain meaningful control over how cultural materials are represented, circulated, and transformed [5, 6]. At the same time, HCI work on explainable and collaborative AI suggests that users need provenance, feedback, and interpretable signals to calibrate trust in generated outputs [2, 3]. Our work connects these threads by treating cultural fidelity as an interface problem: readers need support for noticing cultural shifts that are not apparent from fluency alone.

2.4 Narrative Fidelity as an HCI Problem

Narrative fidelity differs from conventional text quality because the relevant unit of evaluation is not only the sentence but also the relation among characters, events, settings, and social expectations. In interactive systems, users often judge generated stories through perceived fluency, emotional coherence, and usefulness. These cues are valuable, but they can mask transformations in cultural meaning. For HCI, the problem is therefore not simply to build a better automatic score; it is to decide how cultural-risk signals should be presented to users, when a system should ask for additional source material, and how communities should participate in defining fidelity criteria.

Our approach follows this interactional perspective. The automated framework is not intended to declare a final cultural truth. Instead, it identifies candidate transformations that a user, reviewer, educator, or community moderator may need to inspect. This framing aligns with provenance-oriented and collaborative AI systems that help people read generated text more critically rather than treating model outputs as self-evident artifacts [2, 3].

This perspective is especially important for low-resource narrative traditions, where evaluation resources are limited and cultural knowledge may be distributed across families, teachers, storytellers, and local communities rather than encoded in standardized datasets. A system that appears accurate because it produces fluent English or formal Bengali may still erase situated meanings that are obvious only when the retelling is compared with a source tradition. We therefore treat narrative fidelity as a socio-technical problem: it requires metrics, but it also requires interfaces that make those metrics contestable and useful.

A further challenge is that cultural loss often appears as accumulation rather than a single dramatic error. One missing kinship term may seem minor, one simplified motivation may seem harmless, and one explicit moral may seem helpful. Together, however, these changes can make a folktale less recognizable as part of a particular cultural world. This cumulative quality motivates our focus on profiles of displacement instead of one-off error detection.

This also changes how we understand user responsibility. It would be unrealistic to expect ordinary readers to manually audit

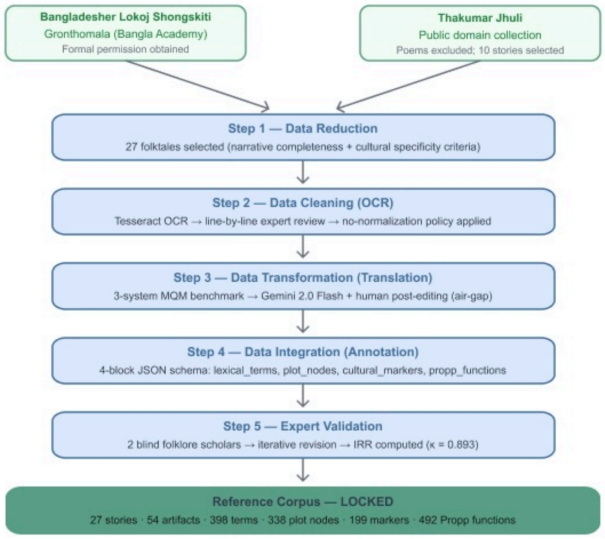


Figure 2: Reference corpus construction pipeline, including story selection, bilingual preparation, expert annotation, and validation.

every generated retelling against a source text, especially when they are using AI because they want help summarizing, translating, or adapting material quickly. At the same time, it would be unsafe for interfaces to treat cultural retelling as a low-risk rewriting task. The design challenge is therefore to create lightweight points of friction that appear when cultural risk is likely, while still allowing legitimate adaptation when users intentionally choose it.

3 Methodology

3.1 Reference Corpus

We constructed a reference corpus of 27 Bengali folktales, yielding 54 Bengali-English story pairs. The corpus was designed to support cultural fidelity evaluation rather than generic semantic similarity. Each story was annotated along four dimensions: culturally non-substitutable lexical items, atomic plot points, socio-cultural behavioral traits, and imposed Western narrative forms. Figure 2 summarizes the corpus construction workflow.

The lexical dimension captures words and phrases whose replacement changes cultural meaning, such as kinship terms, ritual objects, local occupations, and place-specific references. The plot dimension decomposes each story into atomic events so that omissions, additions, and reordering can be measured. The behavioral dimension records culturally situated expectations about obligation, hierarchy, reciprocity, gender, family, and community. The narrative-form dimension captures structural impositions such as explicit moral closure, sanitized characterization, and Western hero-centered arcs.

For each story, annotation began by identifying culturally anchored tokens and phrases that should not be replaced by generic equivalents without justification. Annotators then decomposed the story into plot atoms: minimal narrative events whose removal or reordering changes the story's logic. Next, annotators recorded social and behavioral expectations, such as whether a character's action is meaningful because of age, gender, kinship, class, occupation, or religious and supernatural context. Finally, annotators marked features that would indicate imported narrative form, including the addition of explicit moral summaries, simplified hero-villain oppositions, or endings that resolve ambiguity in a way not present in the source.

This annotation strategy was designed for retelling rather than archival description. The goal was to produce criteria that could be checked against generated outputs. The final corpus contained 398 culturally recognized Bengali lexical terms, 338 atomic plot nodes, 199 socio-cultural markers, and 492 Proppian functions across the 27 source stories. For example, if a retelling keeps a protagonist but removes the kinship relation that motivates the protagonist's action, that change is counted as more than stylistic variation. Similarly, if a retelling keeps a supernatural event but reframes it as a metaphor or dream, the system treats the change as a culturally meaningful displacement rather than a harmless paraphrase.

The bilingual structure of the corpus also allowed us to examine where cultural meaning becomes fragile across languages. Some Bengali terms can be translated directly, but others carry social position, affect, or ritual context that is weakened by a single English equivalent. For these terms, we marked whether a retelling retained the Bengali expression, supplied an explanatory gloss, or replaced it with a generic word. This distinction matters because cultural preservation does not always require leaving a term untranslated; in many educational contexts, a gloss can preserve meaning more effectively than either deletion or unexplained transliteration.

3.2 Generation Design

We generated 1,296 AI-retold stories using six LLMs and four prompt strategies. The model set covered systems from Meta AI, OpenAI, Alibaba Cloud, DeepSeek, Google DeepMind, and Mistral AI: Llama, GPT-OSS, Qwen, DeepSeek, Gemini, and Devstral variants. Prompt strategies varied along two dimensions: language of interaction and source-text anchoring. This enabled us to compare English-mediated versus Bengali-mediated interaction and generic retelling versus prompts that explicitly supplied original source material.

The generation design reflects a realistic HCI concern. Users often ask AI systems to rewrite a story without specifying which cultural details are non-negotiable. Other users provide source text but still rely on the system to decide what should be retained, omitted, or normalized. By varying prompting conditions, we examined whether cultural fidelity improves when the source is visible and whether native-language interaction automatically protects local cultural detail.

The four prompt strategies were selected to reflect common user behaviors. Prompt 1, Simple Control, asked for the story by name only. Prompt 2, Context-Rich, added the collection or regional source. Prompt 3, Constraint-Based, explicitly asked the model to retain culturally specific terminology and moral framing. Prompt 4, Source-Anchored, supplied the complete source story and requested a retelling in the model's own words. Comparing these conditions

allowed us to separate failures caused by missing source information from failures caused by the model's tendency to normalize or simplify culture even when relevant information is available.

Generated outputs were evaluated at the story level and the dimension level. Story-level scores make it possible to compare models and prompts, while dimension-level scores reveal the type of cultural loss. This distinction is important for design: an interface warning that "this retelling may be culturally inaccurate" is less actionable than one that indicates whether the risk concerns lexical substitution, omitted plot atoms, altered social behavior, or structural displacement.

We also treated prompting as an interaction design variable rather than only an experimental control. The same model can behave differently depending on whether a user asks for a "retelling," a "summary," a "child-friendly version," or a "faithful adaptation." These everyday instructions implicitly authorize different degrees of transformation. In our prompt set, the constraint-based and source-anchored prompts were therefore designed to test whether explicit user intent can counteract the model's tendency to smooth culturally specific material. This framing is important for HCI because many deployed systems do not ask users to specify preservation goals before generating cultural content.

To reduce order and formatting effects, generated stories were normalized before scoring. We removed boilerplate openings when they were clearly system-added, preserved Bengali terms when models supplied them, and treated explanatory glosses as partial retention rather than deletion. We did not penalize harmless paraphrase or stylistic modernization when the relevant cultural relation remained intact. This distinction allowed the audit to focus on culturally meaningful displacement instead of rewarding literal copying. It also reflects the practical reality that useful retellings may change surface wording while still preserving cultural structure.

The scoring procedure therefore separated three cases that are often collapsed in automatic evaluation. The first is faithful variation, where wording changes but the cultural relation remains intact. The second is recoverable loss, where a term or event is missing but can be restored from the source annotation. The third is structural replacement, where the output introduces a different explanation for why the story unfolds as it does. This separation helped us interpret whether a model needed better prompting, richer source grounding, or stronger review before the output could be treated as culturally faithful.

3.3 Cultural Fidelity Perception Framework

We compared three automated evaluation designs. Design A measured lexical retention, Design B used embedding-based similarity, and Design C used a rubric-based cultural audit aligned with the four corpus annotations. Design C became the Cultural Fidelity Perception Framework because it detected cultural displacement that remained invisible to surface metrics. The framework is summarized in Figure 3.

A central component is the Structural Displacement Indicator (SDI). SDI targets cases where a retelling preserves broad plot similarity but changes the underlying story grammar. For example, a model may retain the main characters and sequence of events

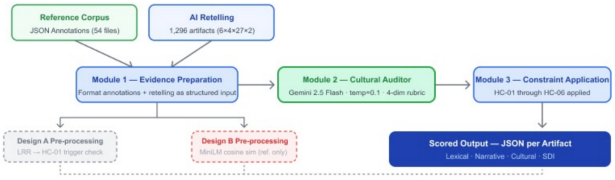


Figure 3: Cultural fidelity assessment pipeline used to score lexical, structural, behavioral, and narrative-form displacement.

while adding a moral lesson, reframing collective action as individual self-realization, or replacing ambiguous supernatural agency with rationalized causality. Such outputs can score highly under semantic similarity metrics while failing culturally.

Design C scores each retelling through a multi-criteria rubric. Lexical fidelity asks whether non-substitutable cultural terms are retained, translated with explanation, or replaced by generic equivalents. Plot fidelity asks whether atomic events remain present and ordered in a way that preserves causal logic. Behavioral fidelity asks whether characters act within the same socio-cultural frame as the source. Narrative-form fidelity asks whether the retelling adds external story grammar that changes the genre or moral structure. These dimensions are combined to produce a cultural fidelity profile rather than a single opaque score.

SDI complements the rubric by focusing on a recurring failure pattern: the output looks like the same story at a distance, but the underlying structure has shifted. A high-SDI retelling may be easy to read and difficult for a non-expert to reject. It may also be attractive in educational or entertainment settings because the story becomes more familiar to global audiences. For this reason, SDI is especially relevant to HCI: it identifies outputs that may require user-facing friction, review prompts, or provenance display.

3.4 Operationalizing Cultural Fidelity

The framework treats cultural fidelity as a profile rather than a binary label. This is important because retellings often mix preservation and loss. A model may retain culturally marked names but remove the social relationship that explains why a character behaves in a particular way. Another model may preserve the sequence of events but change the ending so that uncertainty becomes explicit moral instruction. A single score can help rank outputs, but the profile is what supports interpretation and repair.

For interface design, this profile can be mapped to different forms of feedback. Lexical loss can be shown as highlighted terms that were omitted or generalized. Plot loss can be shown as missing or reordered events. Behavioral loss can be shown as altered relationships or motivations. Narrative-form displacement can be shown as a warning that the retelling introduced a genre convention not present in the source. These feedback types are intentionally concrete: they give users something to inspect rather than asking them to trust an abstract cultural score. They also support different repair actions. A missing term can be restored, a plot atom can be reinserted, a changed motivation can be rewritten, and an imposed ending can be removed or explicitly marked as an adaptation. By

connecting each metric to an editable part of the text, the framework becomes useful for authoring rather than only post-hoc evaluation.

The distinction between preservation and repair also matters. In some settings, a user may want to adapt a story for a new audience and may intentionally change length, vocabulary, or tone. The problem is not that every change is unacceptable. The problem is that current systems rarely make culturally meaningful changes visible. A culturally accountable tool should therefore support informed transformation, not only strict preservation. The audit framework can help identify which changes are stylistic and which changes alter culturally significant meaning.

3.5 Human Verification Study

The human verification study recruited 76 Bengali participants. Participants read story pairs and identified which version was authentic or AI-retold. The study included high-fidelity retellings, where automated scores indicated minimal cultural loss, and low-fidelity retellings, where automated scores indicated major cultural distortion. Participants also reported confidence, allowing us to examine not only accuracy but calibration.

The study design tested whether human readers can function as unaided cultural auditors. If readers reliably detect cultural erasure, automated support may be less necessary. If they struggle when outputs are fluent, then HCI systems need to scaffold attention toward cultural cues that are not captured by style or readability.

Participants were not asked to evaluate grammar or literary quality in isolation. Instead, they were asked to make an authenticity judgment: which version better reflected the culturally faithful story? This task mirrors real-world encounters with AI-mediated heritage content, where users may need to decide whether a generated retelling is suitable for teaching, sharing, archiving, or creative reuse. Confidence ratings were collected because a system design problem emerges when users are both inaccurate and confident. In that case, interface interventions must address calibration as well as accuracy.

4 Results

4.1 Automated Audit Findings

The audit showed systematic cultural compression across models and prompts. Cultural fidelity was low across all six models under English Prompts 1–3. Lexical retention averaged 0.1337, and SDI indicated substantial structural displacement. Prompt 4 improved fidelity most strongly: lexical retention increased from 0.114 in the Simple Control condition to 0.640 in the Source-Anchored condition, while mean SDI decreased from 2.84 to 0.55. Generic prompting produced stories that were smoother, shorter, and more globally familiar than the source. Culturally marked lexical items were generalized, social obligations were simplified, and ambiguous endings were turned into explicit moral conclusions. These changes were not uniformly visible through lexical retention or embedding similarity.

Table 1 summarizes the three evaluation designs. Design A was useful for detecting obvious lexical substitutions but missed cases where a story retained cultural terms while changing deeper structure. Design B overestimated fidelity because two stories can remain semantically similar while differing in culturally important ways.

Table 1: Comparison of automated evaluation designs.

Design	Signal	Main limitation or strength
A	Lexical retention	Misses structural change
B	Embedding similarity	Overestimates fidelity
C	Cultural audit	Detects cultural loss

Design C produced the most useful audit because it incorporated the annotated dimensions of the reference corpus.

Language of interaction produced a notable asymmetry. Bengali-mediated interaction did not automatically improve cultural preservation. Lexical retention dropped by 64.2% when moving from English-mediated to Bengali-mediated interaction under comparable conditions, suggesting that low-resource language use may expose weaknesses in model representations. Source-text anchoring was the most consistently beneficial prompting strategy, but its gains varied across languages and dimensions.

The strongest failures were not always the most visible. Some retellings removed culturally marked details but remained easy to read. Others changed the narrative frame by adding explicit lessons about kindness, courage, or honesty, even when the source story worked through irony, trickery, or unresolved consequence. These changes illustrate why cultural erasure should be understood as transformation rather than deletion alone. The story is still present, but it has been made to conform to a different set of expectations.

The comparison across prompt strategies suggests that user instruction matters, but it does not fully solve the problem. Prompts that requested cultural preservation improved some scores, yet models still made substitutions that appeared to reflect default assumptions about what a good story should look like. Source-text anchoring helped because it reduced the model's need to infer missing details. However, even source-anchored prompts did not eliminate structural displacement, indicating that preservation instructions must be paired with evaluation and review.

4.2 Patterns of Cultural Compression

The audit revealed several recurring patterns. The first was lexical generalization: culturally specific words were replaced by broad English or Bengali equivalents that carried less social or ritual meaning. The second was relational flattening: kinship, hierarchy, and obligation were turned into generic friendship, kindness, or individual preference. The third was causal rationalization: events involving supernatural or cosmological agency were rewritten as coincidence, metaphor, or personal choice. The fourth was moral over-specification: stories that originally relied on ambiguity, irony, or consequence were given explicit lessons.

These patterns are consequential because they change how a story teaches cultural knowledge. Folktales do not only communicate plot; they communicate what kinds of actions are intelligible within a community. When a model replaces relational obligation with individual motivation, it changes the moral world of the story. When it adds a universal moral at the end, it changes how readers are invited to interpret the narrative. Such transformations can make a story easier to consume while weakening its value as cultural transmission.

The model comparison also suggests that cultural erasure should not be framed as a failure of one particular system. Although outputs differed in style and degree of preservation, the tendency toward compression appeared across the model set. This points to a broader issue in generative modeling: systems trained for helpful, fluent, broadly acceptable text may default toward familiar narrative conventions when they encounter culturally specific material. Prompting can reduce the problem, but evaluation and interface design remain necessary.

A recurring pattern across outputs was that culturally specific details were most likely to disappear when they were not required for a simplified plot summary. Local objects, forms of address, and social obligations were often treated as decorative rather than structurally meaningful. This explains why source-anchored prompting improved fidelity but did not eliminate displacement: even when the model had access to the source text, it still had to decide which elements were central. For cultural narratives, that decision cannot be safely delegated to fluency-optimized generation alone.

We observed the strongest compression around elements that make sense through social context rather than explicit exposition. Kinship terms, forms of deference, community expectations, and supernatural causality were often preserved only when they were directly named in the prompt or repeated prominently in the source. When these elements appeared as background assumptions, models were more likely to replace them with generic interpersonal motivation. This pattern matters because much cultural knowledge in folktales is implicit: the story often assumes that readers understand why a younger character obeys an elder, why a promise matters, or why a supernatural encounter has moral force.

The results also show why culturally faithful generation cannot be solved by simply asking for longer outputs. Some longer retellings retained more events, but also introduced additional explanation, psychological motivation, and explicit moral framing. In those cases, extra length increased readability while also increasing narrative-form displacement. This suggests that preservation-oriented prompts should ask not only for detail, but for source-grounded detail: the system should expand what is culturally present rather than adding familiar narrative material from outside the source.

4.3 Human Detection Findings

Participants were not reliable detectors of high-fidelity AI-generated retellings under blind-reading conditions. When AI outputs preserved enough cultural structure to appear plausible, detection accuracy was 49.6%, effectively chance-level. Participants nevertheless reported moderate confidence, indicating an overconfidence risk: fluency and plausibility can make culturally transformed text feel authentic.

Detection improved when cultural distortion was severe. In the low-fidelity condition, accuracy rose to 83.8%, and calibration improved, with mean Brier score decreasing from 0.040 to 0.008. When participants were provided with dynamically generated evaluation feedback from the automated pipeline, calibrated detection accuracy reached 86.0%. Figure 4 summarizes the contrast between blind and scaffolded detection conditions.

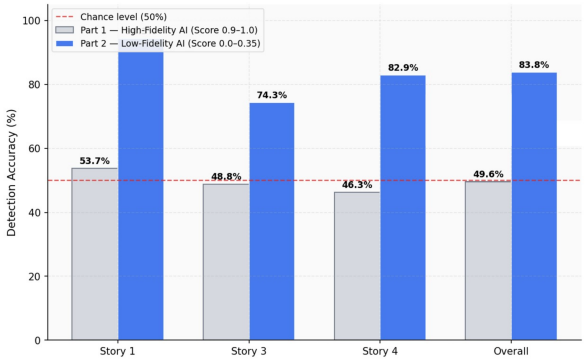


Figure 4: Human detection accuracy across high-fidelity, low-fidelity, and feedback-supported conditions.

The automated and human results aligned in a meaningful direction: as cultural fidelity decreased, human detection increased. The alignment analysis reports a Spearman rank correlation of $\rho = -1.00$ ($p < 0.01$) across the tested story conditions, indicating that lower automated fidelity scores corresponded to higher human detection rates. This does not mean automated scores should replace community judgment. Rather, the results suggest that automated cultural audits can serve as cognitive scaffolds, helping readers attend to cues that are easy to miss when a story is fluent.

Participants' written explanations also showed that readers often relied on surface cues when making authenticity judgments. Some described authentic-looking outputs as "natural," "complete," or "well written," even when the automated audit identified omitted cultural terms or altered narrative closure. Others flagged outputs as AI-generated because they seemed too polished, too explanatory, or too similar to a moral story for children. These explanations suggest that readers were not simply guessing; they were applying reasonable but incomplete heuristics. Fluency, polish, and coherence helped participants make decisions, but these cues did not consistently track cultural fidelity.

The high-fidelity condition is the most important from a safety and design perspective. Participants' near-chance performance does not mean the retellings were culturally empty; rather, it indicates that when a model preserves enough cultural signal, readers may not notice remaining distortions. This creates a threshold effect. Severe distortions are easier to reject, while subtle distortions can pass through human review because they do not violate expectations strongly enough to trigger scrutiny.

The feedback-supported condition demonstrates a possible path forward. Automated feedback did not replace participants' judgment; it redirected attention. When participants saw cues about missing lexical items, altered plot atoms, or structural displacement, they became better able to evaluate the retelling. This supports the design of mixed-initiative cultural review tools in which automated systems surface evidence and humans make contextual decisions.

Importantly, the feedback was most useful when it pointed to specific evidence rather than presenting a generic warning. Participants could reason about a missing term, a changed event, or a newly added moral more easily than they could reason about a

single authenticity score. This suggests that cultural-risk interfaces should be designed around inspectable traces: what changed, where it changed, and why the change may matter for a particular reading context.

4.4 Implications for Reader Calibration

The participant results suggest that cultural expertise and reading fluency do not automatically produce calibrated detection. Readers may recognize Bengali cultural material and still miss the specific ways a retelling has shifted. This is partly because AI-generated text can preserve enough cues to satisfy a general sense of authenticity. It is also because many distortions are relational: the problem is not a single false word, but a changed connection among events, motives, and cultural expectations.

Calibration is therefore a central design target. A tool that merely warns users that a text may be AI-generated would not address the problem observed here. Participants needed evidence about how the retelling differed from the reference. The feedback-supported condition indicates that readers can use such evidence effectively. In practice, this suggests that cultural review interfaces should combine automated highlighting with opportunities for user judgment, annotation, and correction.

The findings also complicate assumptions about community detection. It may be tempting to assume that members of a culture will always identify distortions in their own stories. Our results show that this assumption is unsafe when distortions are subtle and outputs are fluent. In the low-fidelity condition, even participants who reported low familiarity with Bengali folklore could detect obvious distortions, suggesting that severe AI artifacts can be visible through general storytelling cues. Community expertise remains essential, but interface support can help direct that expertise toward otherwise hidden transformations.

5 Discussion

5.1 Why Fluency Is an Unsafe Proxy for Cultural Fidelity

The findings show that fluent retellings can still be culturally unfaithful. This creates a design challenge for generative systems because many current interfaces foreground fluency, helpfulness, grammar, or readability. For culturally sensitive generation, those signals are insufficient. Interfaces should separate linguistic polish from cultural fidelity and should make visible which culturally significant elements were preserved, altered, or removed.

A practical interface could present a cultural fidelity panel alongside the generated retelling. The panel might show retained cultural terms, omitted plot atoms, and detected SDI risks. It might also compare the retelling to source annotations and ask the user whether substitutions are acceptable for the intended context. For example, a classroom adaptation for young children may allow some simplification, while an archive or cultural education setting may require stricter preservation. The key design principle is that the system should not silently decide which cultural changes matter.

These findings also suggest that evaluation should be localized. A general-purpose "cultural sensitivity" score is unlikely to capture the specific forms of displacement that matter for Bengali folktales. Communities need tools that allow them to define non-substitutable terms, narrative boundaries, and acceptable transformations. This points toward participatory dataset construction and review workflows rather than one-size-fits-all benchmarks.

5.2 Six HCI Design Implications

First, cultural fidelity should be displayed as a distinct quality signal rather than folded into generic output quality. Second, systems should expose language-of-interaction asymmetries because prompting in a local language does not guarantee better cultural representation. Third, source-text anchoring should be the default for culturally sensitive retelling, translation, and summarization tasks. Fourth, interfaces should introduce calibrated friction when SDI-like structural displacement is detected, for example by prompting users to review altered plot logic or moral framing. Fifth, community validation loops should be supported so that cultural authenticity is not defined solely by model-internal judgments or external evaluators. Sixth, scalable cultural audits should include structural displacement detection because lexical overlap and semantic similarity can overstate fidelity.

These implications apply at different layers of system design. At the model-interaction layer, prompts and defaults should encourage source grounding. At the evaluation layer, systems should assess multiple dimensions rather than reporting a single confidence value. At the interface layer, users should be shown interpretable evidence of cultural changes. At the governance layer, communities should be able to contest, revise, and extend the criteria used to define fidelity. Without these layers, cultural review risks becoming a cosmetic feature rather than a meaningful safeguard.

A concrete implementation could combine these layers in a retelling interface. Before generation, the system could ask whether the user wants a faithful retelling, a classroom adaptation, a translation, or a creative remix. During generation, it could preserve annotated terms and plot atoms as constraints. After generation, it could present a short cultural fidelity report that separates retained elements from changed elements. If SDI risk is high, the interface could require the user to acknowledge that the output is an adaptation rather than a faithful version. Such a workflow would not prevent creative reuse, but it would make cultural transformation visible and accountable.

5.3 Designing for Use Context

Different use contexts require different thresholds for intervention. In casual creative writing, a user may tolerate adaptation if the system clearly labels the output as transformed. In educational settings, stronger source grounding may be required because students may treat the retelling as representative cultural knowledge. In archival or heritage settings, even small substitutions may require review because the goal is preservation rather than adaptation. A culturally accountable interface should therefore allow fidelity requirements to be configured by context.

One practical design is a layered review workflow. The first layer gives a quick summary of overall cultural fidelity and SDI risk. The second layer lets users inspect evidence by dimension, such as missing terms or altered plot atoms. The third layer supports collaborative review, where community members can accept, reject, or explain transformations. This layered approach avoids

overwhelming casual users while still supporting expert review when the stakes are high.

The same principle applies to model evaluation dashboards. Reporting a mean score across all stories can hide systematic weaknesses on particular dimensions. A model that performs well on lexical retention may still impose Western narrative closure. A model that preserves plot atoms may still flatten social behavior. Evaluation interfaces should therefore expose dimension-level performance and allow reviewers to inspect representative failures.

5.4 From Audit Scores to Interface Actions

A useful cultural audit should lead to concrete interface actions. Low-risk retellings may only need a short provenance note identifying the source and the prompt used to produce the output. Medium-risk retellings may require inline highlights showing terms or events that changed. High-risk retellings, especially those with strong SDI signals, should trigger review before the text is shared as culturally faithful. This tiered approach avoids treating every transformation as equally problematic while still preventing silent cultural replacement.

The audit can also support repair. When the framework detects lexical loss, the system can suggest restoring the original term or adding a gloss rather than replacing it. When it detects plot loss, the system can show the omitted event and ask whether it should be reinserted. When it detects behavioral displacement, the system can ask the user to review motivation and relationship changes. When it detects narrative-form displacement, the system can identify added moral closure or genre shifts. In each case, the goal is not automatic correction but accountable editing.

For authors, educators, and cultural workers, such tools could make AI retelling more transparent. A writer adapting a folktale could see which changes are stylistic and which alter cultural structure. A teacher could compare a generated classroom version with the reference story before assigning it. A community archive could flag generated summaries that require human review. These scenarios illustrate why cultural fidelity is not only an evaluation metric but also a workflow requirement.

5.5 Limitations and Scope

The study focuses on Bengali folktales and should not be treated as a universal model of cultural fidelity. The annotation framework is transferable, but each cultural domain requires community-specific criteria. The human study measures detection in controlled reading conditions; real-world encounters with AI-generated cultural content may involve repeated exposure, social sharing, and additional contextual cues. The model ecosystem also changes rapidly. Because the study includes six contemporary LLMs and four prompt strategies, the contribution is not a permanent ranking of systems but an evaluation approach and empirical demonstration of failure modes that can be retested as models change.

The study also has limitations related to annotation and participant context. Cultural fidelity judgments may vary across region, class, generation, religion, diaspora experience, and familiarity with oral storytelling. A transformation that one reviewer sees as unacceptable erasure may be seen by another as a legitimate adaptation for a new audience. Our framework is designed to surface such transformations, not to erase disagreement. Future deployments should therefore treat the audit as an entry point for discussion and review rather than as a final authority.

Future work should examine longitudinal exposure, participatory annotation with originating communities, and interface prototypes that test how cultural audit feedback changes user trust, workload, and decision-making across educators, storytellers, children, diaspora communities, and professional translators.

5.6 Future Work

Future work should evaluate cultural fidelity tools in live authoring environments. The present study shows that feedback can improve detection, but it does not yet show how users would act on feedback while drafting, translating, or teaching. Interface studies could compare document-level warnings, inline annotations, side-by-side source comparisons, and collaborative review queues. These studies should measure accuracy, trust, workload, perceived agency, and willingness to revise generated text.

A second direction is participatory expansion of the reference corpus. The current corpus demonstrates the feasibility of a structured cultural audit, but cultural authority should not rest on a single annotation process. Folklorists, storytellers, teachers, and community members may disagree about which transformations are acceptable in different contexts. Future systems should support multiple annotation layers and make disagreement visible rather than forcing a single definition of authenticity. This is especially important for oral traditions, where variation is part of cultural practice.

A third direction is cross-cultural comparison. Bengali folktales provide one culturally rich domain, but other low-resource traditions may exhibit different forms of displacement. Some may be more vulnerable to lexical loss, while others may be more vulnerable to changes in ritual sequence, ecological knowledge, or collective memory. Applying the framework across communities would help distinguish general design principles from culture-specific criteria.

Finally, future work should examine how cultural fidelity interacts with language learning, diaspora identity, and children's media. AI-retold folktales may become entry points through which younger or diasporic readers encounter heritage narratives. In those settings, fluency and accessibility are valuable, but they should not come at the cost of replacing the cultural world of the story. Longitudinal studies could examine whether source-grounded feedback helps readers learn culturally specific terms and relationships over time, rather than only detecting whether a single output is authentic.

6 Conclusion

This paper examined cultural erasure in AI-retold Bengali folktales through a corpus of 27 stories, 1,296 generated retellings, and a 76-participant human verification study. The results show that contemporary LLMs can preserve surface coherence while compressing culturally significant lexical, structural, behavioral, and narrative-form features. Conventional lexical and embedding-based metrics missed important forms of displacement, motivating the Cultural Fidelity Perception Framework and SDI as tools for evaluating culturally situated retellings.

The human study shows that readers are not always reliable unaided detectors of cultural erasure. Participants performed near-chance when AI-generated retellings were fluent and high-fidelity, but detection improved when distortions were severe or when automated feedback scaffolded attention to cultural cues. These findings position cultural fidelity as an HCI problem: systems that mediate cultural heritage should not rely on fluency as evidence of trustworthiness. Instead, they should provide source-grounded and community-accountable review mechanisms for identifying what has been retained, transformed, or erased. More broadly, cultural accountability in generative AI requires systems that recognize culture as structure, relation, and context while helping communities sustain their own narrative heritage.

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